

- Integrates a 12V to 5V at 4A **buck converter** with a 5V to 3.3V LDO, **ESP32-C3**, and **SX1281 RF transceiver**, supporting power passthrough for two 12V propellers and six PWM servo outputs at 5V on a board size of 30.5x30.5mm.
- Features a **pi-network matching circuit** and **impedance-matched routing** for the ESP32-C3's power and antenna nets.
- Addresses possible transient voltage spikes from the **source impedance** of a 3S battery and XT60 connector, as well as **resonant frequencies** in **high-Q** capacitors (MLCCs), using **LTspice** simulations.
- Built to be deployed for a fleet of foam fixed-wing UAVs for flight training at the **Waterloo Aerial Robotics Group**.



- Schematic and layout with **Altium Designer**
- PDN analysis with **Power Analyzer by Keysight**
- Frequency domain and time domain-based input filter simulation with **LTspice**, SimSurfing by Murata
- **Impedance-controlled** layout/routing for RF signals

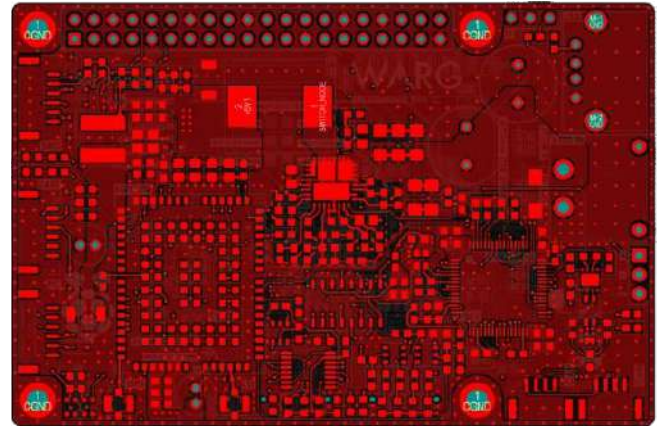
- Splits power to two XT60s, capable of up to **30A** per motor (transient max)
- Matches **ExpressLRS hardware target** for straightforward firmware flashing
- Originally used ESP8285, redesigned with **ESP32-C3** for **stronger PWM signalling**, **6+** free GPIO pins, and improved ExpressLRS compatibility



RPi Interface Rev. C: Layout Design



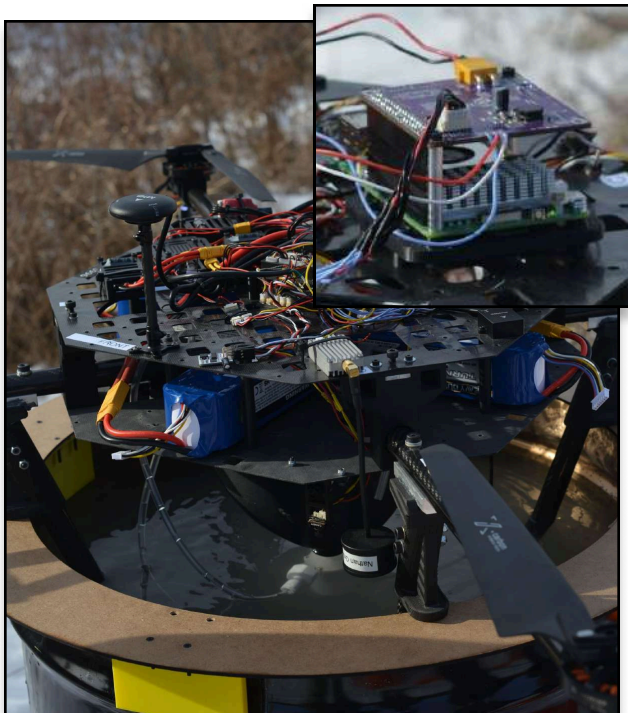
Top 3D View



L1 (SIG/PWR)

Description

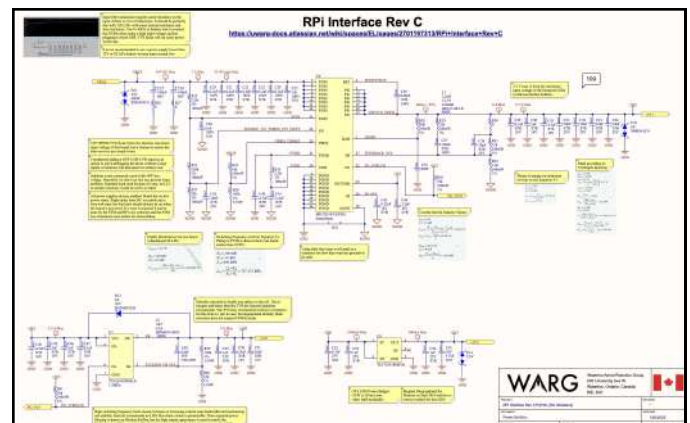
- Raspberry Pi 5 HAT with **STM32F415** and **EG915Q** LTE modem for redundant **command & control** during UAV flight.
- Interfaces with **USB 2.0 FS**, **CAN 2.0**, **SPI** (to Pi), and **UART** (one ExpressLRS & one Pixhawk FC connection).
- Includes MOSFET-based **logic-level shifters** to facilitate communication between 3.3V STM32 and 1.8V EG915 modem.
- Powers Pi & itself using a single **XT30** input connector (input range from 12 to 54V & 12S LiHv tolerant).
- Includes a hardware kill-switch to the Pi to engage **manual override**, which can be triggered over UART from the FC.
- Compact 4-layer, **mixed-signal design** with 2 buck converters, 2 LDOs, ESD protection, and a **guard ring** to chassis ground.



'Pegasus 2' Water Transport Testing
(RPi Interface Rev. B pictured in purple soldermask)

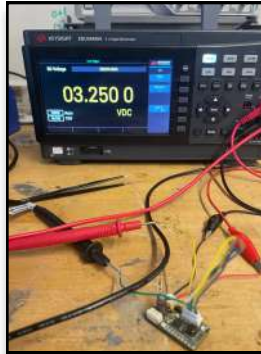
System Integration

- RPi 5 runs local computer vision for autonomous flight.
- Powers RPi through **one XT30**, reducing harnessing.
 - Rev. C eliminates the need for an external 48-12V BEC by including a 12S LiHv-tolerant buck converter.
- Modem was selected for better NA LTE **band coverage**, with lower frequencies prioritized for longer range.



Single Servo Driver: Assembly & Validation

- Assembled **Single Servo Driver** PCBA with solder paste/stencil and hotplate
- Reworked **QFN IC** package with iron, flux, hot air station
- Tested for shorts with DMM probing
- Ran **efficiency tests** by stepping e-load
- Measured **input transient response** to source impedance with an **oscilloscope**

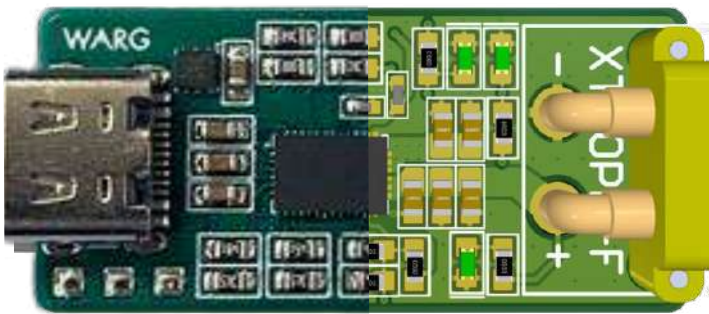


'Single Servo Driver'

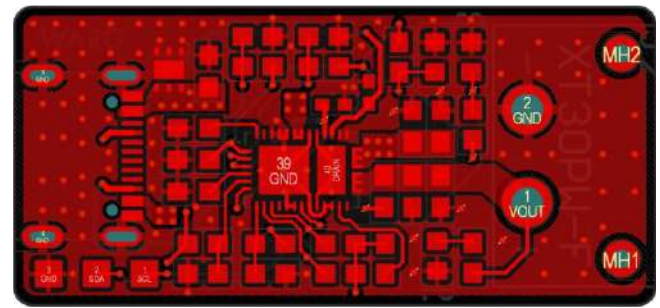


Efficiency Validation

USB-C PD Sink



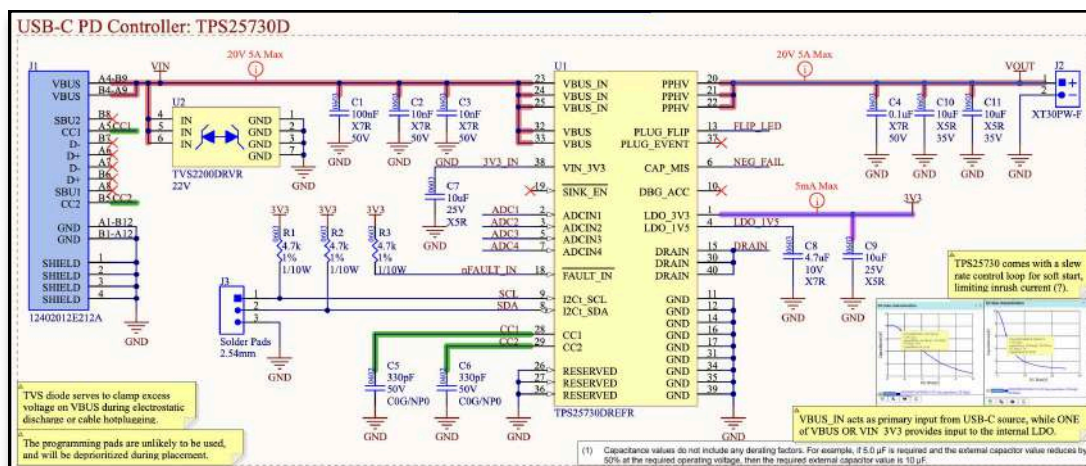
Top View



Top Layer

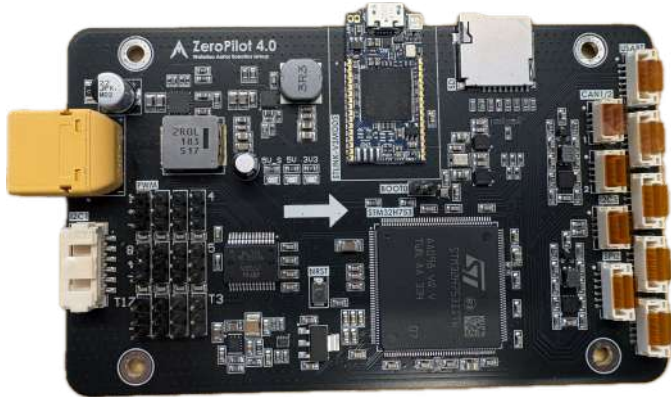
Description

- USB-C to XT30 adapter to deliver up to **100W** (20V 5A) to a drone during stationary debugging and testing.
- PD arbitration** handled by the **TPS25730**, power configuration with resistor dividers on the PMIC's ADC pins.
- Auto-run**; excludes the use of a microcontroller or dedicated buck/boost converters to save on BOM cost.

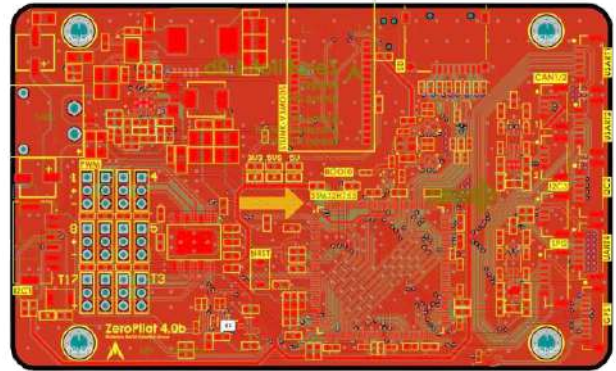


TPS25730 Schematic

ZeroPilot 4.0



Top View

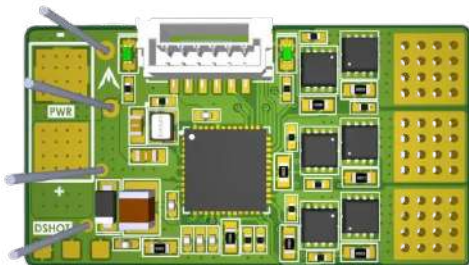


2D View (L1/L3/L4)

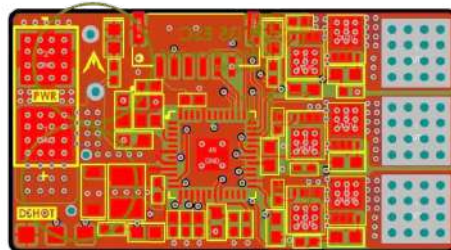
Description

- Fourth iteration of the Waterloo Aerial Robotics Group's fully custom **flight controller** PCBA.
- STM32H753, 2x redundant ICM-42688-P IMUs, an ICP-20100 barometer, maintains compatibility with **ArduPilot**
- Eight 5V servo outputs powered by an isolated 10A buck converter, and a power connector for use with a **PM02D**.
- Features **SD card** slot for log dumps and plenty of CAN/I2C/SPI/UART I/O, with integrated STLINKv3 for easy flashing.
- 3S to 6S battery voltages supported with input **TVS protection** and mix of high and low ESL/ESR input capacitors.
- Single-sided, 4-layer 1.6mm stackup with optimized routing for BOM cost savings, easy manual assembly, and rework.

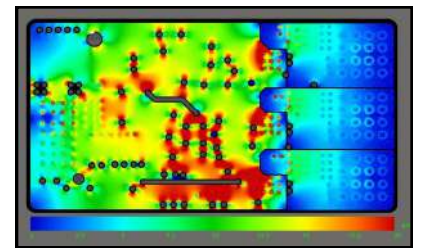
3S 20A ESC



Top 3D View



2D View (L1-L6)

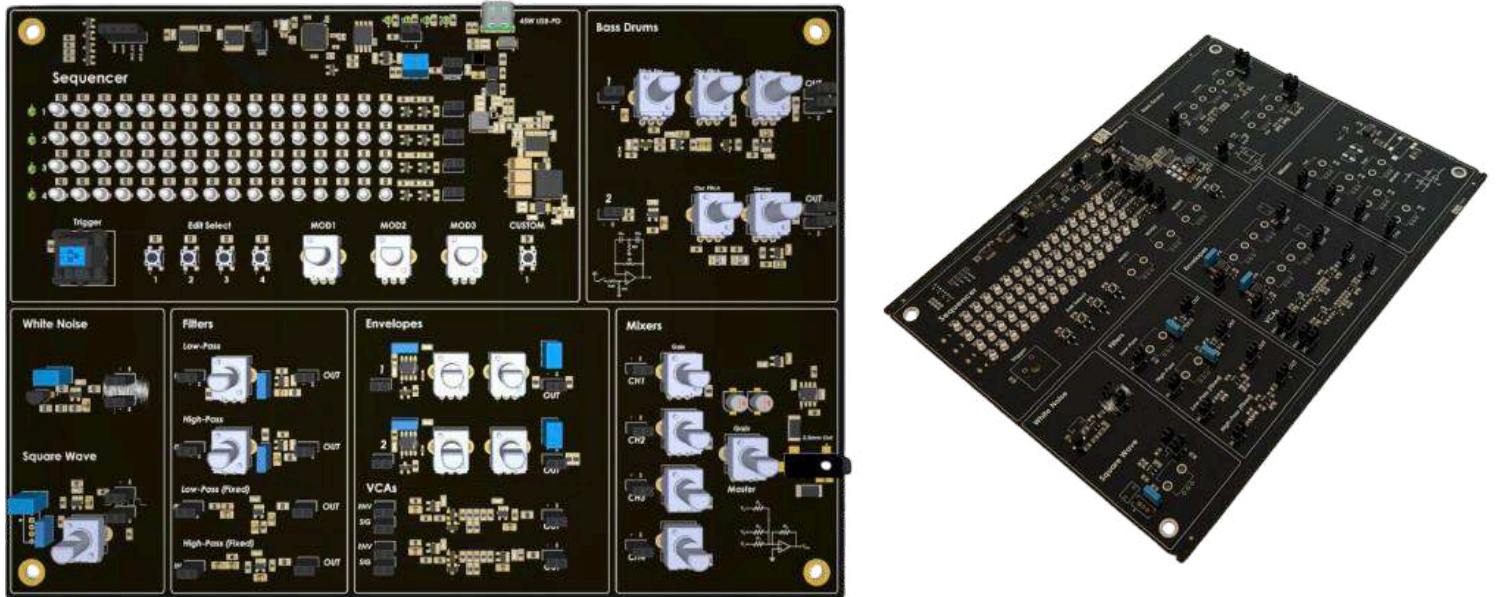


PDN Analysis - (L3/VBAT)

Description

- BLDC motor controller compatible with systems using 3S LiPo batteries. Designed for 15A nominal / 20A max current.
- Compatible with **AM32** ESC firmware (FOSS) through the STSPIN32F0, which includes a STM32 MCU, 12-5V BEC, and gate driver in one 5 mm x 5 mm QFN package. This enables an ultra-compact board size (40 mm x 20 mm).
- Detailed MOSFET selection matrix, power loss calculations, PDN analysis, etc. This ESC uses BEMF sensing.

Analog Synthesizer: Drum Machine



Description

- Analog drum machine based on circuits from the **Roland TR-808/909** synthesizer (and others).
- Analog sig-gen with various VCOs, filters, 555-timer-based envelope, transistor-based VCA, and 3.5mm auxiliary output.
- **4-track, 4-bar** LED sequencer with track editing via pushbuttons fully programmable through RP2040/SN74HCT595.
- Includes I2C interface for optional 1" OLED display and **oscilloscope input** to visualize output waveform.
- Powered by a **single USB-C** port (PD source only). 4-layer, 1.6mm design at around 240x170mm total size.
- Heavily inspired by online DIY synth communities and Moritz Klein's designs!

